

Generation of Tunable Stochastic Sequences Using the Insulator–Metal Transition

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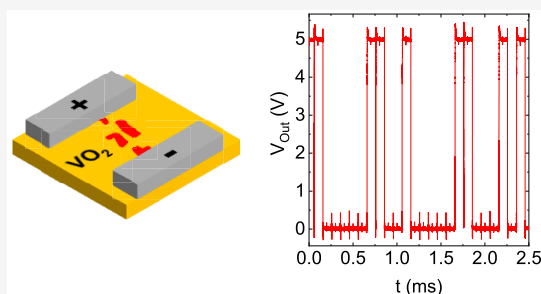
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Supporting Information

ABSTRACT: Probabilistic computing is a paradigm in which data are not represented by stable bits, but rather by the probability of a metastable bit to be in a particular state. The development of this technology has been hindered by the availability of hardware capable of generating stochastic and tunable sequences of “1s” and “0s”. The options are currently limited to complex CMOS circuitry and, recently, magnetic tunnel junctions. Here, we demonstrate that metal–insulator transitions can also be used for this purpose. We use an electrical pump/probe protocol and take advantage of the stochastic relaxation dynamics in VO₂ to induce random metallization events. A simple latch circuit converts the metallization sequence into a random stream of 1s and 0s. The resetting pulse in between probes decorrelates successive events, providing a true stochastic digital sequence.

KEYWORDS: probabilistic computing, random number generation, p-bits, metal–insulator transition, resistive switching



The increasing demand for computational power is fueling the emergence of alternative computation paradigms, beyond the standard von Neumann architecture.¹ One of them is probabilistic computing, where information is represented as the probability of a metastable system (a probabilistic bit or p-bit) to be in a particular state.^{2,3} Initially proposed by von Neumann in 1956,⁴ probabilistic computing gained traction through the 1960s and 1970s because of its promise of massive circuitry reduction,^{5,6} but interest in it eventually faded due to the miniaturization of electronic components and the absence of practical sources of probabilistic bits. In recent years, the growing challenges of big data have motivated a resurgence in probabilistic processing approaches, such as probabilistic programming,⁷ stochastic neural networks,^{8,9} and Boltzmann machines.^{10,11} These methods’ working principles resemble classical counterparts of quantum computing schemes, and they might offer an intermediate platform, easier to realize, between classical and quantum computing.¹²

The development of probabilistic hardware has traditionally been limited by the absence of adequate p-bit sources, that is, scalable hardware capable of producing stochastic and tunable sequences of “1s” and “0s”. CMOS-based solutions, such as linear-feedback shift registers, are large and often not truly stochastic.¹³ Recently, magnetic tunnel junctions near the superparamagnetic limit have shown great promise for fast and scalable implementation of p-bits,^{12,14–16} motivating the search for other systems where intrinsic fluctuations¹⁷ could also be exploited. Natural candidates are materials featuring insulator–metal transitions (IMTs),¹⁸ as they could, a priori, fluctuate between high and low resistance states. For instance, thermal fluctuations in NbO₂ nanodevices have been shown to be

useful in solving optimization problems in an analog way,^{19,20} and the jitter in the self-oscillations of VO₂ devices has successfully been utilized to generate nontunable random numbers.²¹ But to date, no method for generating digital, tunable p-bits using the IMT has been proposed.

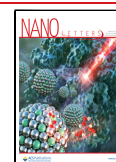
Here, we show that p-bits can be generated using the voltage-triggered IMT in VO₂ nanodevices. We apply two trains of voltage pulses -controlled by two clocks- to our system: the “pumping” and the “probing” trains. The pumping pulses have higher amplitude and always metallize the VO₂. The probing pulses have lower amplitude, sometimes triggering the transition (“1”) and sometimes not (“0”). The triggering probability depends on the probing pulse amplitude, and the stochasticity of the process is ensured due to the randomness of the relaxation path the system follows after the pumping pulse. By using a simple latch circuit, it is possible to generate a continuous string of 0s and 1s with tunable probability. Analysis of the output string indicates true stochasticity with cryptographic quality.

Our devices are fabricated using a 100 nm thick VO₂ film grown by reactive sputtering on top of an R-cut sapphire substrate. Two closely spaced metallic electrodes (Pt or Ti/Au) have been patterned on top of it using electron beam

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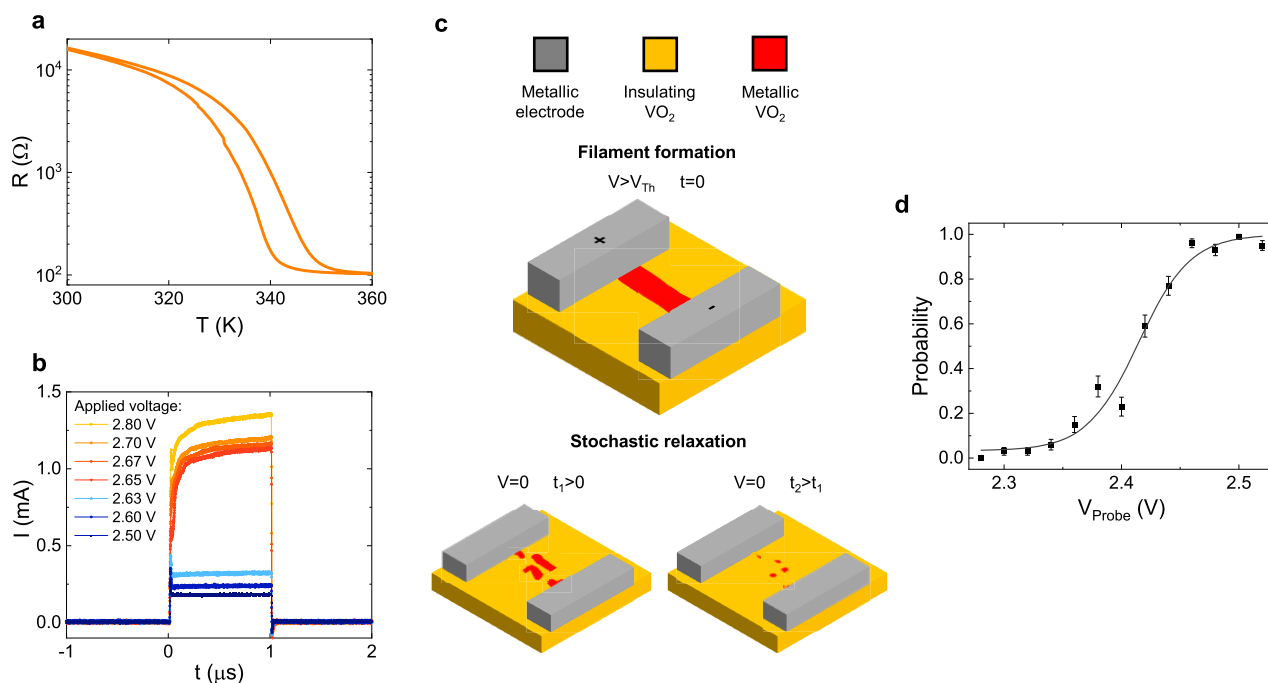


Figure 1. (a) Resistance versus temperature in a VO₂ nanodevice. (b) Current versus time when 1 μ s voltage pulses of different amplitudes are applied. V_{Th} is around 2.64 V. Separation between pulses is more than 1 s to allow for complete relaxation. (c) Schematic representation of filament formation when a voltage is applied (top panel) and its relaxation when the voltage is returned to zero (bottom panel). (d) Probability of inducing the IMT when probing pulses of different amplitudes ($V_{Probe} < V_{Th}$) are sent to a VO₂ nanodevice 100 μ s after the transition has been triggered by a pumping pulse ($V_{Pump} = 3 V_{Th}$). Measurements were done at room temperature ($T = 300$ K).

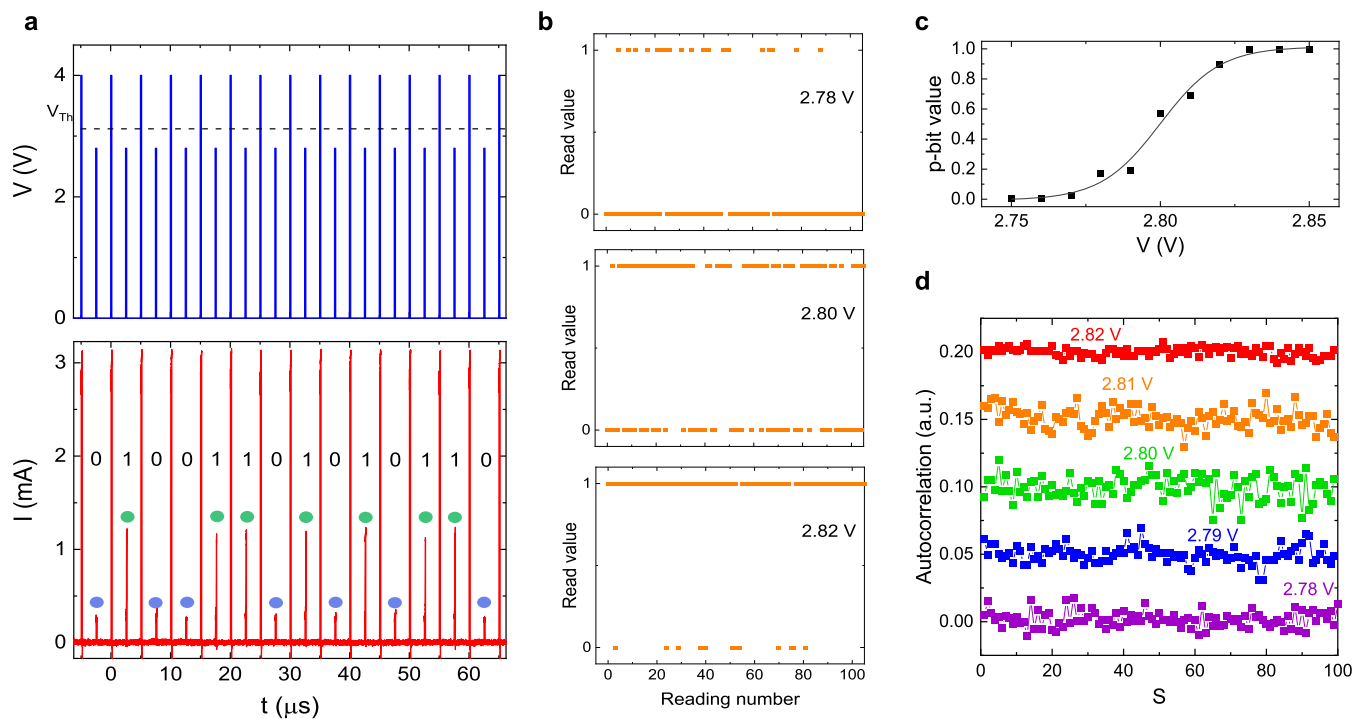


Figure 2. (a) (Top panel) Voltage versus time in a sequence of equally spaced pumping and probing pulses applied to a VO₂ nanodevice at $T = 300$ K. Horizontal dashed line indicates V_{Th} . Separation between pumping (or probing) pulses is 5 μ s. (Bottom panel) Corresponding current versus time measurement. Green dots mark events in which a probing pulse induces the IMT (1), while blue dots indicate when it does not (0). (b) Sequence of read 0s and 1s generated with the protocol shown in (a). Three cases are shown, corresponding to $V_{Probe} = 2.78, 2.80$, and 2.82 V. $T = 300$ K. (c) p-bit value, or ratio of 1s to 0s as a function of V_{Probe} . (d) Autocorrelation versus sequence shift, S , for different values of V_{Probe} . To improve clarity, autocorrelation data corresponding to 2.79, 2.80, 2.81, and 2.82 V was shifted vertically by 0.05, 0.10, 0.15 and 0.20, respectively. Pumping pulse separation was 5 μ s and $T = 300$ K.

lithography (see methods in the [Supporting Information](#)). Electrodes are 400 nm wide, are separated from each other by a 100–300 nm gap and form low resistance ohmic contacts with the VO₂ film. [Figure 1a](#) shows resistance versus temperature in one such device. A sharp, first-order IMT is observed with a resistance drop of around 2 orders of magnitude. This transition can also be induced electrically by applying a large enough voltage.²² The voltage-driven IMT holds much promise for application in related fields such as spiking neural networks^{23–25} or optoelectronics.^{26,27} [Figure 1b](#) shows current versus time when different voltage pulses are applied to a device which is initially insulating ($T = 300$ K in all experiments). For low pulse amplitudes, little current flows through the device as it remains in the insulating state. But once a threshold voltage (V_{Th}) is surpassed, partial metallization takes place and the current rapidly increases. We will refer to these voltage pulses above V_{Th} as pumping pulses, since they always trigger the IMT.

The voltage-triggered IMT is known to happen via percolation of a metallic filament between the metallic electrodes,^{28,29} as schematically depicted in the top panel of [Figure 1c](#). Once the voltage returns to zero, the device locally cools down and starts relaxing back into the insulating state, as depicted in the bottom panels of [Figure 1c](#). The first order character of the IMT in VO₂ plays a crucial role in this process: it has been recently shown that the metastability of the metallic phase leads to slow relaxation dynamics, which can be orders of magnitude slower than the cool down time.³⁰ This has important consequences in the transport properties. If a second pulse, this time below V_{Th} , is sent to the device before it relaxes completely, there is a nonzero probability of triggering the IMT again. We will refer to these pulses below V_{Th} as probing pulses, and they may or may not induce the transition. [Figure 1d](#) shows the probability that a probing pulse of amplitude V_{Probe} induces the IMT 100 μ s after it has been triggered by a pumping pulse ($V_{Pump} = 3 V_{Th}$). The probability has a sigmoidal dependence on V_{Probe} , varying smoothly from 0 to 1. Whether a specific probing pulse triggers the IMT depends on the specific arrangement of unrelaxed metallic islands within the device at the moment the pulse is applied. Since the relaxation path followed by the system after each pumping pulse is variable,^{22,30} cycle-to-cycle stochasticity is expected.

The top panel in [Figure 2a](#) shows the protocol we propose for implementing p-bits using this phenomenology. Two intercalated trains of voltage pulses are applied to a VO₂ nanodevice: a fixed amplitude ($>V_{Th}$) pumping train and a tunable amplitude ($<V_{Th}$) probing train. In our proof-of-concept experiment, we use a function generator to regulate the probing amplitude, but a gated transistor could also be employed for this task. The bottom panel in [Figure 2a](#) shows current versus time in the same device. As expected, the IMT is induced by every pumping pulse, leading to a sharp increase in current. When a probing pulse is applied, the IMT is induced in some cases (green dots), which are to be considered 1; but not in others (blue dots), which are considered 0. The first order, discontinuous character of the IMT in VO₂, implies that only two well-defined device states (conducting or insulating) will be induced when the probing pulse is applied, naturally producing a digital output. [Figure 2b](#) shows three strings of 1s and 0s obtained in this way, for three different amplitudes of the probing pulse. Lower voltage pulses yield strings that are mostly 0s, while higher voltages create mostly strings of 1s. This can be better seen in [Figure 2c](#), where the 1s to 0s ratio

(the nominal value of the p-bit) is plotted as a function of V_{Probe} . Our protocol produces tunable p-bits, whose value can be continuously varied from 0 to 1 by controlling the applied voltage.

All data shown in [Figure 2](#) were taken keeping a period of 5 μ s between probing pulses. Similar results were observed for periods of 1, 2, 10, and 20 μ s, as shown in [Figures S1–S10](#). The typical working V_{Probe} range depends on this period with V_{Probe} being lower for shorter periods. This is expected since for short periods there are more remaining metallic islands when the probing pulse is applied. In our experiments, we applied the probing pulse half a period after the pumping pulse for the sake of simplicity. But different time separation between pump and probe could be used with V_{Probe} becoming lower as this time is made shorter.

In order to assess whether the sequence is truly random, we computed the autocorrelation function. It compares how similar the sequence is to itself but with the sequence elements (X_i) shifted by S positions forward in the chain. S can take any integer between 0 and N , the total sequence length. We define the autocorrelation in [eq 1](#) where P is average of the sequence, or p-bit value

$$\text{Autocorrelation}(S) = \frac{1}{(N-S)} \sum_{i=1}^{N-S} (X_i - P) \cdot (X_{i+S} - P) \quad (1)$$

For $S = 0$, [eq 1](#) compares the sequence to itself and thus becomes the variance of the series, σ^2 , which is in general different from zero. A truly random sequence has a zero autocorrelation for any $S \neq 0$. [Figure 2d](#) shows the autocorrelation as a function of S , for different V_{Probe} . As a function of S , the autocorrelation fluctuates around zero within noise levels, indicating a purely stochastic sequence of 1s and 0s. Similar results were observed for periods of 1, 2, 10, and 20 μ s, and are shown in [Figures S1–S10](#).

The stochasticity of our signal can be further tested. For the particular case of $P = 50\%$, our p-bit can behave as a random number generator. We used our device to create a random sequence of 49 000 1s and 0s with $P = 50.00 \pm 0.01\%$. The sequence is long enough to be subjected to 12 of the 15 tests of the NIST Suite for Random Numbers Generators for Cryptographic Applications,³¹ passing all 12 tests. The remaining three tests require sequences over 1 000 000 digits, unfeasible to record with our experimental setup. In order to obtain P as close as possible to 50%, eight independent sequences were combined using XOR logic. Details can be found in the [Supporting Information](#).

While our protocol produces truly stochastic p-bits, the signal output is not convenient since the 1s and 0s are only readable for a short amount of time, during the probing pulse. This can be solved using a latch circuit that holds the transient voltage across the device. In our case, we add a load resistor in series with the device and use a clocked NAND latch, as depicted in [Figure 3a](#). The NAND latch works in inverted logic: the output goes to zero voltage whenever its set input is logic 0 and the reset is logic 1. Since it is a clocked latch, the output is only changed when a voltage is simultaneously applied to the clock terminal. For operation, we apply both pumping and probing pulses to the VO₂ nanodevice through the load resistor ($R_{Load} = 500 \Omega$ in our case). The voltage V_{In} between load and VO₂ is applied to the set terminal of the latch, while the inverted logic value of V_{In} is sent to the reset terminal using a NOT gate. The probing pulses that are sent to

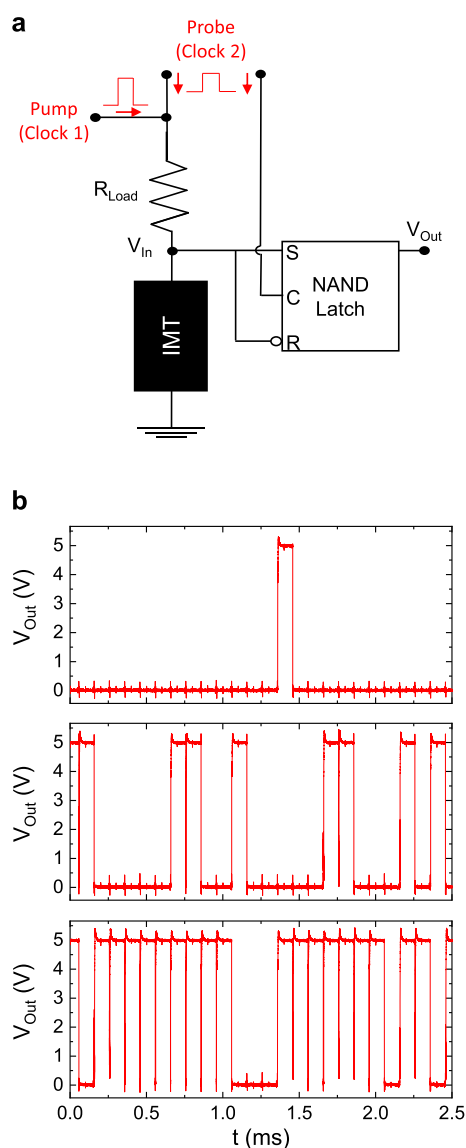


Figure 3. (a) Schematic representation of the latch circuit described in the main text. R_{Load} and the VO_2 device act as a voltage divider, meaning V_{In} is high whenever a voltage is applied and the VO_2 is insulating and decreases once the VO_2 becomes metallic, inverting the set and reset terminal values in the NAND latch. The output is only modified while the probing pulse is enabling the clock terminal. (b) Output voltage versus time for the circuit represented in (a). $T = 300$ K, $V_{\text{Pump}} = 3.4$ V and pump pulse separation is $100 \mu\text{s}$. The three panels correspond to V_{Probe} of 2.94 V (top), 2.95 V (middle), and 2.96 V (bottom).

the device are also applied to the clock terminal of the latch. If the probing pulse does not induce the IMT, V_{In} is high, driving set to 1, reset to 0, and the output to 0. If the IMT is triggered, the situation is reversed and the output changes to 1. Since the output can only be updated during the probing pulse, the last value is held in between pulses. This can be observed in Figure 3b for three different probing pulse amplitudes. The output voltage stays mostly around 0 V, around 5 V, or fluctuates between them depending on the probe. Figure S11 shows the probability as a function of V_{Probe} as well as the autocorrelation functions for several cases, confirming that adding these components to the setup does not compromise the stochasticity of the p-bits.

This method for generating p-bits is scalable, because both the VO_2 devices and the few gates needed to implement a latch can be produced at the submicrometer scale. This would potentially allow for patterning very large-scale arrays of p-bits as part of a chip architecture. All p-bits could rely on the same two system clocks for creating the pumping and probing pulses, and probing voltages could be individually adjusted using a transistor. Regarding speed, here we demonstrate operation up to 1 MHz, but are constrained by device size and limits of the measuring setup, making it difficult to use voltage pulses shorter than 100 ns. Smaller devices or different geometries, such as a vertical configuration, could dramatically speed up the process to values closer to the gigahertz. Switching speeds below 1 ns have been demonstrated in materials such as VO_2 and V_2O_3 .^{32,33} We must also mention possible drawbacks of our approach. The most salient might be the narrow V_{Probe} range (~ 0.1 V) that allows varying P between 0 and 1. This problem, also present in magnetic tunnel junctions, might pose a challenge for properly controlling p-bit values in large integrated circuits. Temperature changes or device-to-device variability could induce small V_{Probe} variations that would result in large P changes. For instance, we analyzed over 10 VO_2 nanodevices, finding similar phenomenology among them but a relatively large threshold voltage variability $\Delta V_{\text{Th}} \approx 0.3$ V, larger than the working voltage range. However, these problems can be tackled: device-to-device variations can be greatly reduced with further optimization, and temperature sensitivity can be minimized exploring materials with a nonthermal IMT, such as GaTa_4Se_8 .³⁴ Another potential source of variability might be device degradation over time due to repeated switching. Similar devices to the ones presented here showed no measurable changes in the threshold voltage after 10^8 switching cycles,²² whereas other VO_2 works have reported no degradation in the material properties after 10^{10} cycles³⁵ or after 100 h of continuous operation.³⁶ However, stability over months and years would have to be monitored before any technological implementation is possible. These considerations, however, are beyond the proof-of-concept scope of our paper and should be carefully addressed in future works.

In conclusion, we developed a method for generating probabilistic bits using materials with first order metal–insulator transitions. We induce the transition electrically and use the intrinsic stochasticity of the relaxation pathway as a source of randomness when retriggering the IMT. By combining VO_2 nanodevices with a latch circuit we create tunable and truly stochastic p-bits which can be potentially downscaled below the micrometer scale. Our work presents metal–insulator transitions as a new physical system where applications for probabilistic computing can be explored, contributing to the prospects created by recent progress in the field.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.1c04404>.

Details on sample fabrication, randomness evaluation using the NIST suite, performance for different pulse periods (Figures S1–S10) and performance with the latch circuit (Figure S11) (PDF)

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Notes

The authors declare no competing financial interest.

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